Frequency dependence in EOM theories and its relation to the BSE

I. EQUATION-OF-MOTION THEORIES

We start by analyzing the frequency dependence in Equation-of-Motion (EOM) theories. Neutral excited-states within a EOM approach are determined from

$$\langle\Psi_0|\hat{C}_\mu^\dagger[\hat{H},\hat{R}_n]|\Psi_0\rangle = \omega_n\langle\Psi_0|\hat{C}_\mu^\dagger\hat{R}_n|\Psi_0\rangle, \tag{1}$$

with $\hat{R}_n$ denoting the excitation operator for the n-th excited state

$$\hat{R}_n = \sum_\mu r_\mu^{(n)}\hat{C}_\mu. \tag{2}$$

Eq. (1) can be written in matrix form as

$$\mathbf{H}\mathbf{r}_n = \omega_n\mathbf{r}_n. \tag{3}$$

Assuming the excitation operator to be restricted to single and double excitations, i.e.,

$$\hat{R}_n = \sum_{ia} r_i^{a,(n)}\hat{a}_a^\dagger\hat{a}_i + \frac{1}{4}\sum_{ijab} r_{ij}^{ab,(n)}\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_j\hat{a}_i, \tag{4}$$

where i,j (a,b) label occupied (virtual) orbitals. The block-structure of the Hamiltonian matrix $\mathbf{H}$ in the basis of single and double excitations is then given by

$$\mathbf{H} = \begin{pmatrix}\mathbf{H}_{\mathrm{ss}} & \mathbf{H}_{\mathrm{sd}}\\ \mathbf{H}_{\mathrm{ds}} & \mathbf{H}_{\mathrm{dd}}\end{pmatrix}, \tag{5}$$

where the subscripts 's' and 'd' refer to single and double excitations, respectively. Downfolding the Hamiltonian in the space of single excitations leads to the frequency-dependent effective Hamiltonian

$$\mathbf{H}_{\mathrm{eff}}(\omega) = \mathbf{H}_{\mathrm{ss}} + \mathbf{H}_{\mathrm{sd}}(\omega - \mathbf{H}_{\mathrm{dd}})^{-1}\mathbf{H}_{\mathrm{ds}}. \tag{6}$$

For the following analysis we assume a diagonal doubles block, i.e.,

$$\mathbf{H}_{\mathrm{dd}} = \langle\Phi_{ij}^{ab}|\hat{H}|\Phi_{kl}^{cd}\rangle = \delta_{ik}\delta_{jl}\delta_{ac}\delta_{bd}(\epsilon_a + \epsilon_b - \epsilon_i - \epsilon_j) \tag{7}$$

where ϵ_p denote the canonical Hartree–Fock orbital energies. The coupling block read

$$\begin{aligned}\mathbf{H}_{\mathrm{sd}} &= \langle\Phi_i^a|\hat{H}|\Phi_{jk}^{bc}\rangle\\ &= \delta_{ik}\langle ab||jc\rangle - \delta_{ij}\langle ab||kc\rangle\\ &+ \delta_{ab}\langle jk||ic\rangle - \delta_{ac}\langle jk||ib\rangle.\end{aligned} \tag{8}$$

This illustrates that we can distinguish between to fundamental building blocks in the frequency-dependent part of the effective Hamiltonian.

These can be categorized into two types: Genuine two-particle interaction terms

$$\Gamma_{iajb}(\omega) = \sum_{kc}\frac{\langle ab||kc\rangle\langle kc||ij\rangle}{\omega - (\epsilon_c + \epsilon_b - \epsilon_k - \epsilon_i)}, \tag{9}$$

or terms with a spectator particle

$$\begin{aligned}&\delta_{ab}\Sigma_{jl}(\omega')\\ &\qquad= \delta_{ab}\sum_{klc}\frac{\langle jk||ic\rangle\langle ic||kl\rangle}{\omega - (\epsilon_c + \epsilon_b - \epsilon_i - \epsilon_k)}\\ &\qquad= \delta_{ab}\Sigma_{jl}(\omega - \epsilon_b).\end{aligned} \tag{10}$$

The latter can be directly identified as self-energy-like term dressing the single-particle propagator.

With this, we can rewrite the frequency-dependent effective Hamiltonian as a sum of genuine two-particle interaction terms and disconnected one-particle contributions

$$\begin{aligned}\mathbf{H}_{\mathrm{eff}}(\omega)_{ijab} &= \delta_{ij}\left(\delta_{ab}\epsilon_a + \Sigma_{ab}(\omega + \epsilon_j)\right)\\ &- \delta_{ab}\left(\delta_{ij}\epsilon_i + \Sigma_{ij}(\omega - \epsilon_b)\right) + V_{iajb} + \Gamma_{iajb}(\omega).\end{aligned} \tag{11}$$

This is just for illustration purposes. The signs might be wrong.

Summarizing the frequency dependence as $\nu = \omega - \epsilon_b$ and $\nu' = \omega - \epsilon_j$, we can rewrite the effective Hamiltonian in terms of these shifted frequencies

$$\begin{aligned}\mathbf{H}_{\mathrm{eff}}(\omega,\nu,\nu')_{ijab} &= \delta_{ij}\left(\delta_{ab}\epsilon_a + \Sigma_{ab}(\nu')\right)\\ &- \delta_{ab}\left(\delta_{ij}\epsilon_i + \Sigma_{ij}(\nu)\right) + V_{iajb} + \Gamma_{iajb}(\omega).\end{aligned} \tag{12}$$

Interpretation: The diagrammatic rules for constructing the BSE, i.e., via the Parquet approximation, do not allow for the inclusion of the self-energy insertions directly, due to two-particle reducible and irreducible contributions being separated from one-particle contributions to avoid double counting from the usage of a interacting one-particle Green's function. The fermionic frequencies therefore reflect the fact that the evaluation of the self-energy has to be shifted by the corresponding bosonic frequency compared to the single-particle treatment via the Dyson equation.

Next: Comparison with the multifrequency kernels derived by Titou. Furthermore, the derivation above decouples excitations from deexcitations. I conjecture that this is the reason why the particle/hole self-energy insertions appear only for 2h1p/2p1h intermediate states. In the CC case, one would have additional static 2p1h/2h1p contributions arising from the contraction with the T_2 amplitudes.